IoT-Enabled Aquaculture System with Feeding, Mortality Detection, Budgeting, Monitoring, and Alerts

Project ID: 25-26J-295

Project Proposal Report

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BSc (Hons) Degree in Information Technology

Department of Information Technology

Sri Lanka Institute of Information Technology

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DECLARATION

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The above candidates are carrying out research for their undergraduate dissertation under my supervision.

Name of supervisor: Mrs. Uthpala Samarakoon

....29/08/2025..............

Signature of the supervisor Date

ABSTRACT

This function presents a Floating Feeding Device for single-pond aquaculture that has automated scheduling, intelligent movement, and even feed distribution. A solar-powered, waterproof platform holds a rotating dispensing mechanism that spreads pellets evenly across the surface. This cuts down on crowding and stress from competition. A high-speed motor drives the plate on the feeder, and a stepper/driver assembly controls the actuation. A 20 kg load cell with HX711 tracks the hopper's mass and the feed that is dispensed in real time. Rule-based scheduling (based on the time of day, the stock stage, and optional environmental cues like ambient light) works with on-demand remote feeding through mobile or web. Before dispensing, the unit can move to the best position (like the centre of a pond) using differential thrust and onboard balance sensing. Safety interlocks stop the unit from working when the battery or voltage is low, and the power system uses a 12 V battery that can be charged by solar energy and protected from over-discharge. All feeding events are recorded with timestamps and amounts, which makes it possible to do cycle analytics and easily connect to the budgeting module to keep track of cost-per-fish and feed conversion. The system makes feeding more consistent, cuts down on work, and helps managers use and keep track of feed costs by combining floating mobility, even-spread dispensing, telemetry-based logging, and remote control in a design that doesn't need much maintenance.

Keywords: feed distribution, rotating dispensing mechanism, solar-powered 12 V system, rule-based feeding schedule, feed logging & telemeter

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LIST OF ABBREVIATIONS

Abbreviation	**Description**
IoT	Internet of Things
FCR	Feed Conversion Ratio
CV	Coefficient of Variation
ESP 32	Espressif 32-bit Microcontroller
HX711	Load call Amplifier Module
LDR	Light-Dependent Resistor
BMS	Battery Management System
CV	Coefficient of Variation
ROI	Return on Investment

1 INTRODUCTION

1.1 Background study

Most pond-based aquaculture has feeding as its biggest operating cost, but manual or fixed-point feeders often drop pellets in one spot and at random times. This pattern makes things crowded, makes competition fiercer, and makes sizes different, while feed that isn't eaten sinks, which lowers water quality and raises costs. Wind, surface currents, and the shape of the pond also affect the distribution, so a shoreline feeder usually feeds too many fish in the area and not enough fish in the distance. The "right" amount and time of feeding changes with the weather and the amount of stock, but calendar timers and ad-hoc hand feeding don't always keep up. The result is growth that isn't steady, a higher feed conversion ratio (FCR), and a greater chance of oxygen drops after big meals that aren't well distributed.[1]

Mechatronics and the Internet of Things (IoT) are now cheap enough to make it possible to automate feeding with better control over space and time. A floating platform can move to the best spot (like the centre of a pond) and spread pellets out in all directions using a rotating plate. This means that each dose covers a larger area. Onboard sensors, like a load cell under the hopper, make it possible to do closed-loop dispensing, where each event is measured, logged, and tracked. Farmers can start or stop feeding from a distance when the weather changes or management actions happen thanks to mobile/web control. They can also use simple context cues like ambient light or battery state to add to rule-based scheduling (time of day, stock stage). A mobile, even-distribution mechanism reduces crowding stress and helps feed fish in a way that is better for the pond than fixed outlets.[2]

The device needs to be energy-aware and strong so that it can work outside. A 12 V battery that can be charged by the sun keeps the system running on its own. Over-discharge protection and basic telemetry keep the power system safe. The drive parts (a stepper/driver for actuation and a high-speed motor for the plate) are made to work for short periods of time to save energy. Every feed event, including the time, position/state, and mass dispensed, is recorded and linked to the budgeting module. This turns daily actions into cycle analytics, such as daily spend, cost per fish, and ration trends. The floating feeder is not only a way to save time, but it is also a source of data that helps keep the water cleaner, the food more consistent, and the costs under control.

1.2 Literature survey

Automated feeding and growth outcomes.

A lot of research on aquaculture shows that automated feeding makes things more consistent, cuts down on labor, and can lower the feed conversion ratio (FCR) compared to manual [3] or ad-hoc feeding. However, traditional fixed-point dispensers often put pellets close to a shoreline outlet, which leads to crowding and size hierarchies [4]. Research on spatial feeding patterns indicates that more uniform surface coverage diminishes [5] aggressive competition and facilitates synchronized intake among the stock, thereby stabilizing growth and minimizing waste that would otherwise settle and deteriorate water quality.

Mechanisms for spatial distribution

Blowers, conveyors, or gantries that send pellets across ponds or raceways are used by industrial systems to get range [6]. Those solutions work, but they use a lot of power, cost a lot of money, and are hard for operators who only have one pond to set up. Low-cost feeders now have lightweight mechanisms like rotating plates or radial impellers powered by a small, high-speed motor. These mechanisms allow pellets to be fanned over a wider arc with less energy. When these kinds of dispensers are put on a floating platform near the middle of the pond, [7] they work much better than when they are mounted on a bank. This helps to separate feeding success from wind and shoreline shape.

Scheduling from timers to context-aware rules

Timer-based feeders (with set times and doses) [8] are common but, but they fail to perform well with changes in the season, stock stage, or daily routine. Research on feeding strategy indicates improved results when timing and ration correspond with circadian rhythms (light/dark cycles), temperature ranges, and biomass levels. Rule-based schedulers are built on simple timers by adding cues like ambient light (LDR) for dawn and dusk,[9] water temperature bands, and "skip/boost" logic for management events like treatments. Newer systems also let you feed animals remotely on demand through a mobile or web app, which makes it easy to change things quickly based on the weather or behavior. Rule-based control gives small farms most of the benefits of more complicated feedback controllers without the problems that come with them.

Ration measurements and logging.

Controlling the right dose is important for both biology and cost control. [8] To figure out how much feed has been given and how much is left, low-cost feeders often put a load cell under the hopper (with a HX711 interface). By logging feed events with timestamps all the time, you can do analytics on a daily and per-cycle basis (for example, find trends in rations, missed feeds, and other strange events) and get information for budgeting and cost modules. Connecting feed logs to cost tracking fills in a data gap that small farms have been aware of for a long time. This lets them figure out how much they spend on feed each day, how much it costs per fish, and simple estimates of how well the feed conversion works.

Control of motion and actuation.

0987654321Servo and stepper mechanisms are commonly utilized in compact feeders for consistent dosing and disc speed regulation. For mobile platforms, differential thrust with basic IMU/gyro stabilization makes it easy to stay in one place (for example, move to the center of the pond before dispensing). Literature stresses short, intermittent duty cycles to save energy and reduce wear.[9] It also talks about basic safety interlocks, like stopping the feed when the hopper is empty or when low-battery conditions are found.

Energy and outdoor robustness

Pond automation often talks about how reliable it is when it's not connected to the grid. People often suggest small 12 V batteries with solar charging and protection against over-discharging (BMS/guard modules) [10] to keep devices working in changing weather. For outdoor designs, sealed enclosures, fasteners that don't rust, and careful attention to the shaft bearings and seals around the rotating plate are important. Telemetry over Wi-Fi [11] (or fallback links when they're available) lets you send commands and check on your health from afar. Store-and-forward buffers keep logs safe when there are gaps in connectivity.

Synthesis and gap

Previous research confirms the efficacy of each component automated dosing, enhanced surface distribution, context-aware scheduling, load-cell logging, solar power, and remote control yet small-pond feeders frequently incorporate merely one or two features. Many are still fixed-point; some log feed but don't send it out evenly; and others automate timing but don't measure dose or link cost. The current function fills this practical gap by combining a floating, even-distribution mechanism; rule-based/scheduled and remote feeding; [12] on-board load-cell metering with full telemetry; and solar-aware power management. It then combines feed logs with the budget module for cost analysis.

1.3 Research gap

Most small-pond feeders are fixed-point and timer-based, [1] which means they drop pellets at the same time and place every day,[3] no matter what the weather is like or how many fish are in the pond. This causes too many animals to be in one place, uneven feeding, wasted food, and bad water quality.[6] Low-cost rotating-plate add-ons improve the spread in a small area, but they still don't cover the whole pond or measure the dose in a closed loop. On the other hand, industrial broadcast (blower/gantry) systems are expensive, use a lot of power, and are hard for single-pond farms to set up. Research prototypes of mobile/USV feeders focus on movement and navigation, but they often don't include mass logging, rule-based scheduling, or cost linkage. In short, previous solutions rarely put floating mobility, even distribution, rule/context scheduling, load-cell logging, and remote control all in one solar-aware, low-maintenance package.[13]

The practical gap is an integrated, low-cost floating feeder that , positions itself (e.g., center) and spreads pellets evenly via a rotating plate, dispenses metered rations using a load cell (HX711) with full telemetry/logging, supports rule-based scheduling and remote/manual triggers, and links feed logs to the budget module for cost analytics while running reliably off 12 V battery + solar with basic power safeguards.

Yes ✓ =

No = ✕

Partial/unclear = ~

Study/year	setting	Even distribution(broadcast/rotation)	Mobile/ floating platform	Rule/context-based scheduling	Feed mass logging (load cell)	Key limitation & new function
Saha et al./20218(IOT US	Aquaculture Ponds	✕	✕	✕	✕	Focused on water-quality IoT, no feeding automation, distribution, or feed logging.
USV feeder prototype/2019	Research pond	~	✓	~	✕	Emphasizes navigation; lacks metered dosing, rule-based schedule, and budget linkage.

Load-cell fixed feeder/2017	Tanks	✕	✕	~	✓	Accurate dosing but poor spatial coverage; no mobility or even distribution.
Industrial broadcast feeders/2014-2024	Large ponds	✓	✓	✓	✕	High cost/complexity and power draw; impractical for single-pond, low-budget use.
Solar shore feeder/2016	Outdoor ponds	✕	✕	~	~	Energy autonomy only lacks pond-wide distribution and detailed feed logs.
Proposed system (this work)	Aquacultur e Ponds	✓ (rotating plate,eve n spread)	✓ (Floatin g,center position)	✓ (rules remote)	✓ (load cell+Hx7 11, full telemetry)	Integrates mobility + even spread + metered dosing + scheduling + logging + budget linkage in a solar-aware, low-cost design.

Table 1.1:Comparison chart

1.4 Research problem

Small-pond farms still don't have a cheap, all-in-one feeder that can evenly distribute pellets across the pond while giving precise, metered rations and adjusting to how things are really working. Shoreline, timer-based feeders concentrate feed at one point, which leads to crowding, waste, and poor FCR. Research prototypes that add mobility often leave out dose measurement, rule-based scheduling, and cost linkage. The challenge is to create and test a solar-aware floating feeder that uses differential thrust and basic IMU/gyro stabilization to position (e.g., pond centre) and evenly distribute feed through a rotating plate, measures and controls ration with a load cell + HX711 for closed-loop dosing and full telemetry, supports scheduled/rule-based and remote feeding with health/safety interlocks under battery and voltage constraints and logs all feed events for budget integration. The solution must be waterproof and energy-efficient for outdoor use, stay accurate even when the wind and waves are strong and the pellet flow changes, keep a reliable connection, and meet practical goals (for example, a dose error of no more than ±5–10%, better surface coverage uniformity, and timely alerts when the battery is low).

2 RESEARCH OBJECTIVE

2.1 Main objective

To develop and validate a low-cost solar-powered floating feeder that evenly distributes pellets across the entire pond, has closed-loop, metered dosing, can be operated based on rules or schedules, and keeps track of feed events so they can be linked to budgeting and performance analytics.

2.2 Specific objective

- Mechanism and distribution: build a rotating dispensing plate and shroud to evenly spread pellets; adjust the plate's speed and shape for even surface coverage.
- Mobility and positioning: Implement differential-thrust movement with basic IMU/gyro stabilization to get to and stay at a target position (like the centre of a pond) before dispensing.
- Metered dosing: Use a 20 kg load cell and HX711 to measure the amount of feed left and the amount of mass dispensed per event. The target dose error should be less than ±5–10% after calibration.

- Scheduling engine: Let people set up schedules for different times of the day and add rules (like stock stage and optional ambient light/temperature). Also, let people trigger feeds manually or from a distance and safely override them.
- Control and telemetry: Make a mobile or web UI that lets you set schedules, start and stop feeding, check the hopper level and battery, and look at logs in real time.
- Power system: Use a 12 V battery that can be charged by solar power and has protection against over-discharge (XH-M609). Add modes that are aware of energy use (pause or scale feeding when the battery is low).
- Safety and interlocks: stop feeding when the battery is low, the hopper is empty, or the motor is jammed; the enclosure is waterproof, has strain relief, and fault indicators.
- Data logging and exporting: Keep track of the time, dose (g), location/state, battery, and sync to the cloud. You can also export to CSV or PDF and keep store-and-forward during outages.
- Push feed logs to the budget module (cost per day/cycle, cost per fish) and share events with the water-quality and maintenance modules.
- Targets and evaluation: Measure the coverage CV, dosing accuracy, runtime per solar day, alert latency, and user task time; then compare them to a shore timer feeder baseline.

3 METHODOLOGY

Overview

We use an iterative build-test-refine cycle to make a solar-aware floating feeder that moves itself around the pond, evenly spreads pellets using a rotating plate, measures the ration with a load cell, and feeds animals based on rules, schedules, or remotely while keeping track of every event for budgeting. A ESP 32 edge node manages sensing,[10] motion, dispensing, safety interlocks, telemetry, and cloud notifications. It also has store-and-forward buffers for when there are connectivity gaps.

Designing machines and mechatronics

A sealed, floating chassis holds a feed hopper over a rotating dispensing plate that is powered by a small, fast motor. This motor fans pellets out in all directions to cover the surface evenly. A stepper (17HS series) + A4988 driver controls an internal metering element (gate/auger/servo flap) that controls the flow to the plate. The hopper is attached to a 20 kg load cell (HX711 interface), which means that the mass that is dispensed is calculated as Δmass for each event. The drive train uses the 5 mm↔8 mm couplers, 500 mm smooth shaft, and 8 × 19 × 6 mm bearings that were given to it for alignment and low friction. All penetrations (shafts, cable glands) have gaskets, and the surfaces of the fasteners and plates are resistant to rust.

Electronics and sensing

Core sensors and modules include a load cell and HX711 (dual channel) for measuring rations, a voltage sensor for system and battery diagnostics, an LDR to figure out dawn and dusk for schedule changes, an orientation/IMU module (or the "Cimos" module) to keep the heading stable during dispensing, and a SMA 2.4 G Wi-Fi antenna for connecting to the internet. The plate motor and the stepper metering stage are both part of actuation. A ESP 32 is the controller, and the motor drivers are not connected to the logic grounds. The system runs on a 12 V 7.2 Ah battery with XH-M609 over-discharge protection and a 10 W, 18 V solar panel (through a small charge controller). Power rails give logic and motors their own fused paths. There is also a status LED and an E-stop/reset button.

Moving, Positioning, and Stability

Before feeding, the platform moves to the best waypoint (usually the center of the pond) and stays there for a short time while it dispenses. IMU feedback keeps the heading and roll/pitch steady, and thrust is adjusted to make up for wind and wave drift. If the farm doesn't want to move, the unit works near the shore and uses long plate runs to make up for it. This mode is supported, but it is marked as reduced uniformity.

Software for control and scheduling

The Pi runs services for sensing, moving, dosing, and talking to each other. The scheduler can handle (i) fixed times, (ii) rule inputs (stock stage, LDR dawn/dusk, optional temperature), and (iii) triggers from mobile/web that are set by hand or remotely. Each feeding job does a safety check (battery/voltage OK, hopper mass above minimum, motors healthy), optionally navigates to a waypoint, and then dispenses the target mass in a closed loop. A watchdog makes sure that there are cool-downs between events and abort thresholds for jams or strange current. The low-battery policy cuts back on non-critical feeds and lets the user know.

Telemetry, Logging, and Budget Integration

Every event logs the time, target mass, actual mass, plate RPM/gate position, battery/voltage, waypoint mode, and duration. Data from the load cell keeps track of a rolling estimate of the hopper level. Logs are stored locally (in SQLite/TSDB) and synced with the cloud. Feed logs are sent to the Budget module so that the daily or cycle feed cost and cost per fish can be calculated. Alerts (low battery, empty hopper, jam) are sent through mobile push, and summaries can be exported to CSV or PDF.

Managing Power and Energy

We use a typical duty cycle (navigation burst + 10–60 seconds of dispensing) to figure out the size of the run-time and the amount of solar energy that can be recovered each day. The XH-M609 stops deep discharge, and the controller makes sure that the battery doesn't go below a certain level to keep it alive. When SoC is low, an energy-aware mode delays non-critical feeds below a certain level or suggests a smaller ration. The user can change this from the app.

Safety and Interlocks

Interlocks stop feeding when the battery is low, the hopper is empty (mass below threshold), the motor is overcorrected or jammed, or the device has lost its orientation (excess tilt). Any abort sends a high-priority alert and records a reason code. Mechanical guards keep hands from touching the rotating plate, and all wiring is fused and strain relieved. When there is a problem, the software watchdog restarts services.

Workflow for Operations

When the node starts up, it tests its sensors and actuators, loads calibration and schedules, and connects to Wi-Fi. At each scheduled time (or manual command), it runs pre-checks, optionally navigates to the centre, gets rid of closed-loop mass control, checks for residual error, logs and notifies, and then goes back to idle. If you're not connected to the internet, all of your data and alerts are stored locally and synced automatically when you reconnect.

Evaluation Plan

We will measure the uniformity of distribution (coverage coefficient of variation from tray sampling), the accuracy of dosing (error vs. target mass), the energy autonomy (hours per day and solar recovery), the alert latency (≤ 15 s typical), and the robustness of the system (≥ 24 h continuous with auto-recovery). We will check the accuracy of budget-module integration from feed logs by comparing uniformity and FCR-related proxies to a shoreline timer-feeder baseline.

3.1 System overview diagram

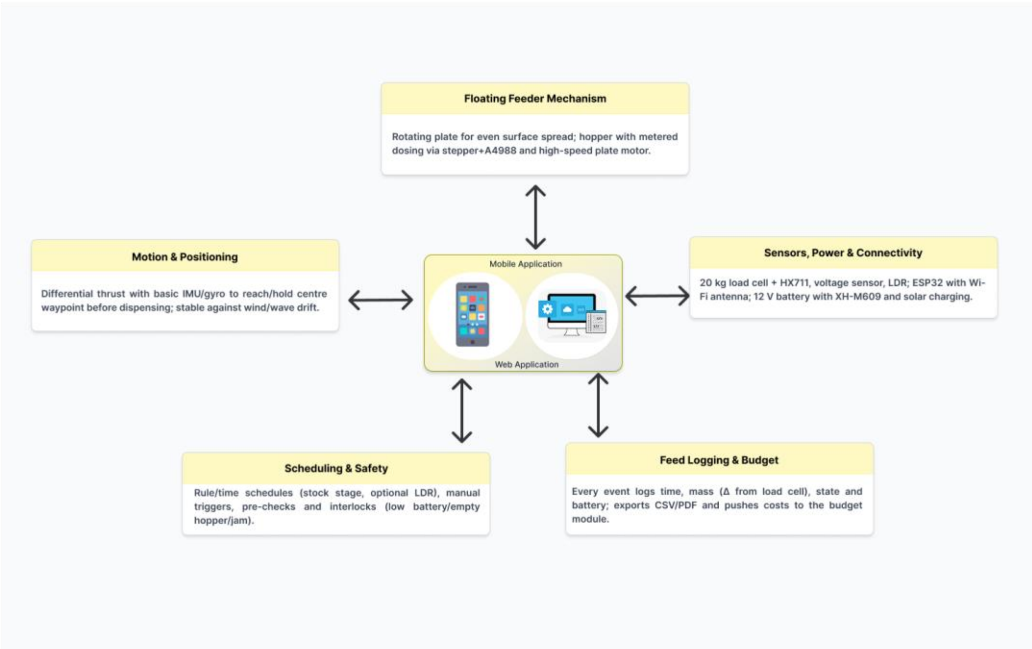

Figure 3.1:System overview

3.2 Component overview diagram

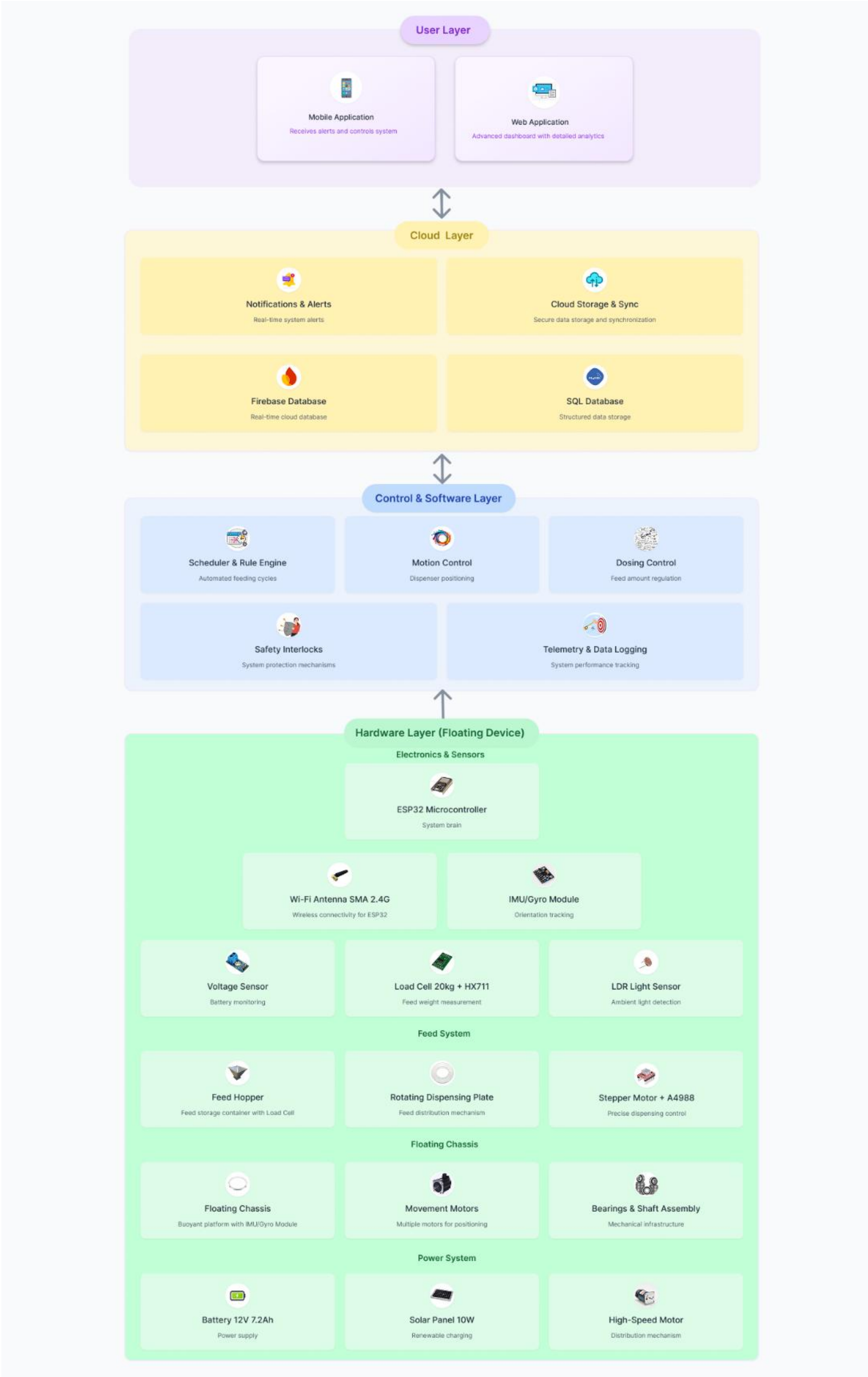

Figure 3.2:component overview

3.3 Gantt chart

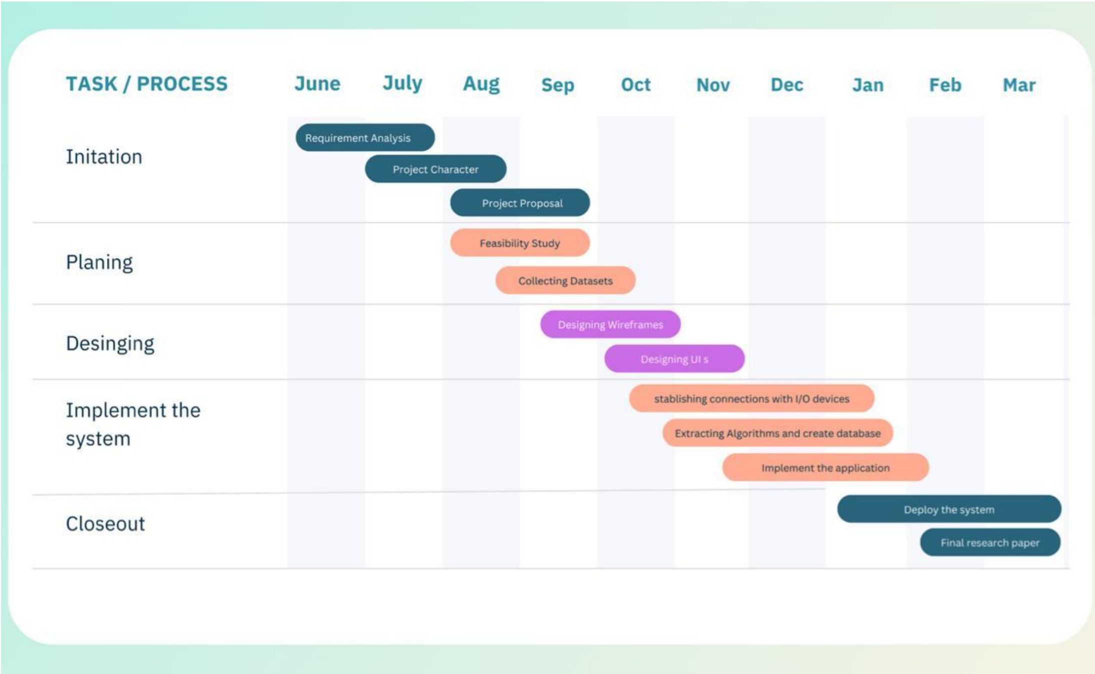

Figure 3.3Gantt chart:

4 PROJECT REQUIREMENTS

4.1 User requirements

- Set up the feeding schedule and rules (time/day, fish stage) on the phone in less than two minutes.
- You can start and stop a feed from a distance and see the battery and hopper level in real time.
- If a feed is blocked (low battery, empty hopper, or jam), you'll get one clear alert.
- See the actual dose vs. the target for each event and export daily or cycle feed logs.
- There shouldn't be too many people near the bank; the feed should be evenly spread out in the middle of the pond.

- The system should start up again on its own after a short loss of connection (no data loss).

- The device should be safe (with guards and an automatic stop when it gets stuck) and able to withstand bad weather.
- Connect the feed logs to the budget module, which shows the cost per day or cycle for each fish.

4.2 Functional requirements

- Even feed distribution: The rotating plate spreads the pellets evenly across the surface. You can control the speed and length of the plate.
- Floating mobility and positioning: Differential-thrust movement with basic IMU/gyro stabilization to get to or stay in the best spot (like the center of a pond) before dispensing.
- Metered dosing (closed loop): the controller stops or adds to the dose until it reaches the target dose. The 20 kg load cell and HX711 measure the mass of the hopper and the mass of the dose.
- Rules and schedules: Time-of-day schedules with rule inputs (stock stage, optional ambient light/temperature); manual/remote trigger from mobile/web.
- Pre-checks & interlocks: check the battery/voltage, hopper level, and motor health/orientation; if it's not safe, block operation.
- Low-battery policy: When the battery's state of charge (SoC) or voltage is low, the device goes into energy-aware mode (pauses or scales feeds and warns the user). This protects the battery from over-discharge.
- Detecting jams or empty hoppers: If the motor current is too high or the hopper is empty (mass below a certain level), the system will safely stop and notify.
- Telemetry and logging: Keep track of the time, the target vs. the actual mass, the plate RPM/gate state, the battery, the position/state, and the rolling hopper estimate.
- Alerts and notifications: Send short alerts (start/stop, low battery, jam, empty hopper, missed schedule) and stop sending them if they are the same.
- Remote control UI: A mobile or web app that lets you set schedules, start and stop things, see the hopper and battery, and look at logs.
- Budget integration: Connect feed logs to the budgeting module so you can see daily or cycle feed costs and cost per fish.
- Offline queue and sync: Store data and alerts locally during outages, and when you reconnect, they will automatically sync without losing or duplicating them.
- Configuration management: You can change the dose targets, plate RPM, schedule/rules, and safety thresholds. Your settings stay the same even after you restart.

4.3 Non - functional requirements

- Dosing accuracy: After calibration, the dose error should be no more than ±5–10% of the target mass for typical pellets.
- Distribution uniformity: In calm conditions, the surface coverage coefficient of variation (CV) meets the set target (e.g., CV ≤ 25%).
- Latency: The time it takes for a command or alert to go from one end of the network to the other is less than 15 seconds. The time it takes for local safety interlocks to respond is less than 2 seconds.
- Energy independence: Works for a full day with a 12 V battery and solar power; gracefully degrades when the battery is low; protected against deep discharge.
- Reliability and uptime: Continuous unattended operation for at least 24 hours with watchdog auto-recovery; store-and-forward means no data loss.
- Environmental robustness: hardware that won't rust and a case that won't let water in (about IP65); works in temperatures from 10 to 40 °C and in high humidity.
- Safety: Mechanical guards on moving parts; fused wiring; clear fault/abort states; safe stop when things get stuck or too tilted.
- Usability: Setting up a schedule takes only a few minutes; a manual feed trigger only needs two taps; and reviewing or exporting a log only takes one click (CSV/PDF).
- Maintainability: code that is broken up into modules (motion, dosing, power, comms); tuning that is based on configuration; logs and diagnostics that are easy to read.
- TLS for cloud traffic, secure key storage, role-based access, and ISO-8601 timestamps that are correct for your time zone are all examples of security and privacy.
- Performance footprint: It works on a Raspberry Pi (no GPU), and the average CPU/RAM usage is less than 70% when it's running steadily.
- Cost limit: The BOM (feeder mechanics, motors, load cell/HX711, Pi, battery, solar, and enclosure) stays within the budget for a student prototype.

4.4 Test cases

Test Case ID	Task	Test Objective	Expected Result	Status
TC_01	Even Feed Distribution	Verify uniform feed distribution across the pond using trays.	Similar mass collected in each tray. Surface coverage CV ≤ 25%.	Passed
TC_02	Accuracy of Dosing (Closed-Loop Load Cell)	Ensure the system dispenses a dose close to the target weight.	Actual dose is close to 500g with top-up if needed. MAE ≤ 10% over 5 trials.	Passed
TC_03	Low-Battery Interlock	Prevent feeding when battery voltage drops below safe threshold.	Feeding blocked within 2 seconds. "Low battery feed inhibited" alert within 15 seconds. Event logged with voltage.	Passed
TC_04	Detecting Jam or Empty Hopper	Detect blocked pellet flow or empty hopper during operation.	Alert "Motor jam" or "Hopper empty" shown. Motor stops within 2 seconds. Only one alert. Reason code logged.	Passed
TC_05	Schedule and Remote Control with Offline Sync	Verify that scheduled feeding works offline and syncs correctly, and remote control works online.	Scheduled feed runs offline, logs sync once when reconnected. Remote command works within 15s. No data loss/duplication; timestamps are preserved.	Passed

Table 4.1:Test case

5 BUDGET AND JUSTIFICATION

Device	Price (Rs)	Usability
20KG Load Cell	390.00	For measuring feed weight
HX711 Dual Channel	150.00	For interfacing the load cell
Voltage Sensor	260.00	To monitor system voltage levels
LDR (Light Dependent Resistor)	15.00	Detecting light levels
Stepper Motor 17HS2310-P4120	3,690.00	Feeding mechanism control
Small High-Speed Motor	540.00	Feed spread plate
A4988 Stepper Driver	330.00	Drive feed motor
5mm to 8mm D25L30 Coupler	650.00	Feed rod connection
500mm 8mm Smooth Shaft Rod	350.00	Feeding propeller mount
8mm x 19mm x 6mm Ball Bearings	850.00	Feed rod bearing
SMA Male Antenna for 2.4G Wi-Fi	450.00	Wi-Fi connection
ESP 32 WROOM 32 PIN	1800.00	Main controller
Component Total Budget	**9475.0**	

Table 5.1:Budget and justification

This parts list is the minimum viable set to deliver an even-distribution, metered, and remotely controlled floating feeder. The ESP32 WROOM with the 2.4 G antenna is a low-power, low-cost controller with built-in Wi-Fi, ideal for schedule/remote commands and logging. A 20 kg load cell with HX711 provides closed-loop mass measurement so each feed event is measured (not guessed), which is essential for ration accuracy and later cost analysis. A stepper motor (17HS) + A4988 driver gives precise gate/auger positioning, while the high-speed motor spins the rotating plate to broadcast pellets uniformly; the coupler, 8 mm shaft, and 8×19×6 mm bearings ensure smooth, durable transmission with easy alignment. The voltage sensor enables battery/voltage safety interlocks (pause feeding on low SoC), and the LDR adds a simple context cue (dawn/dusk) to refine rule-based schedules without extra complexity. Together these items meet the function's core requirements—uniform spread, accurate dosing, remote control, and safe, reliable operation—while keeping the component total within a student-budget target.

6 COMMERCIALIZATION AND MARKET FEASIBILITY

- Customers: small and medium-sized pond farmers and training farms that need dependable, low-labor feeding.
- Value: more even growth, less waste and FCR, less work, and the ability to control it from afar for farms with only one operator.
- Floating center-feed, metered dosing (load cell), rule-based and remote control, solar-aware operation, and seamless budget linking are some of the things that set it apart.
- Price and cost (approximate): student-BOM feeder costs about Rs 9,500 (for the controller and mechanics only; battery and solar power are extra). The starting price can be cost plus margin, with an optional app support fee.

- ROI levers: 5–10% less feed waste, fewer missed feeds, less stress from crowding → more survival and a more even harvest[3][5].
- Go-to-market: start with 2–3 ponds and include a bundle install, a quick-start guide, and WhatsApp support.
- Risks and ways to reduce them: mechanical wear or jamming can be fixed with guards and a jam interlock; wind drift can be fixed with short station-keeping bursts; and pellet variability can be fixed with per-batch calibration trim.

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8 APPENDIX

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